Subham Agrawal

B.Tech, Civil and Infrastructure Engineering
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SUMMARY

Final-year IIT Jodhpur undergraduate with a minor in Mathematical and Computational Economics. Builds statistical models in Python that estimate the risk of rare events and checks them on data they have never seen. Seeking credit risk analyst roles.

EDUCATION

Degree	Institute	Year
B.Tech, Civil and Infrastructure Engineering	Indian Institute of Technology Jodhpur	2023 – 2027
Minor, Mathematical and Computational Economics	Indian Institute of Technology Jodhpur	2026 – 2027

Coursework: Probability, Statistics and Stochastic Processes; Econometrics Lab; Games and Economic Behavior (Grade A); *ongoing:* Financial Derivatives and Risk Management, ML for Economics

PROJECTS

Landslide Susceptibility Mapping using GIS and Machine Learning 2026
B.Tech Project under Prof. Susmita Panda | *Python, scikit-learn, XGBoost, Google Earth Engine* Live app | GitHub

- Built a model that scores the probability of a landslide from 4,431 past events and 14 risk factors, comparing five classifiers.
- Found that standard random testing overstated accuracy (AUC 0.97 vs 0.85 on unseen regions); the final model was tuned on regions it had never seen (AUC 0.905).
- Graded the district into risk zones; the 30 m risk map flags 71.9 km of major roads and ∼57,000 people.

Machine Learning Stock Selection on the Nifty 100 2026
Independent project | *Python, LightGBM, scikit-learn* Live app | GitHub

- Showed that a common validation shortcut inflated a model's Sharpe ratio 7 times (3.68 vs 0.53) and proved the leak with a placebo test on random data.
- Checked 16 years of price data for errors and measured survivorship bias (19.2% vs 9.1% a year for the index).

Volatility Dynamics and the Fear Index: Nifty 50 and India VIX 2026
Team project, Financial Derivatives and Risk Management | *Python, pandas, statsmodels* Live app | GitHub

- Showed that India's fear index (VIX) priced more volatility than the market delivered on 78% of days (2020–26).
- Tested a VIX-timed hedge through the March 2020 crash: it cut a Nifty portfolio's worst loss from −38% to −22%.

Key Factors for Initial Funding of Indian Startups 2025
Research paper with Somya Jangir under Dr. Sanyka Banerjee Report

- Compiled and cleaned data on 585 funded Indian startups (2019–21, USD 1.95B) and built a six-factor framework of what drives early-stage funding.

COMPETITIONS AND CERTIFICATIONS

VC Analyst Case Study: Sattva Frontier (Fin Maverick) 2026
Team, Sandstone fest, IIT Jodhpur Ranked 4th of 159 teams

- Screened 20 startup opportunities and pitched a USD 10M allocation across three picks to a simulated venture capital investment committee.

Inter IIT Tech Meet 13.0 (Zelta Labs), Quant Team Member *built crypto price models; team ranked 11th overall* 2024
Quantitative Finance Cohort'24, Quant Delegate *options pricing, volatility models* (Certificate) 2024

POSITIONS OF RESPONSIBILITY

Assistant Head of Finance, Prometeo'25 *Technical Festival, IIT Jodhpur* 2025

- Ran budgeting and fund allocation for events and artists: about INR 15 lakh of the festival's INR 40 lakh+ budget.

Head of Media, Prometeo'26 *led 25+ juniors; several reels crossed 500k views* 2025 – 2026
Cultural Captain, Inter IIT Cultural Meet 7.0 and 8.0 *led a 25+ member contingent; top-10 finishes* 2024 – 2025

ACHIEVEMENTS

- **JEE Main 2023:** top 1% of 1.3 million candidates. **IOQM 2020:** top 300 in Odisha (Certificate). **CBSE Class X (2020):** 92%.
- **WorldQuant Challenge 2024:** Bronze level (Certificate).

SKILLS

- **Statistics:** classification models, regression, hypothesis tests, out-of-sample validation, drawdown analysis
- **Tools:** Python (pandas, NumPy, statsmodels, scikit-learn), SQL, Excel